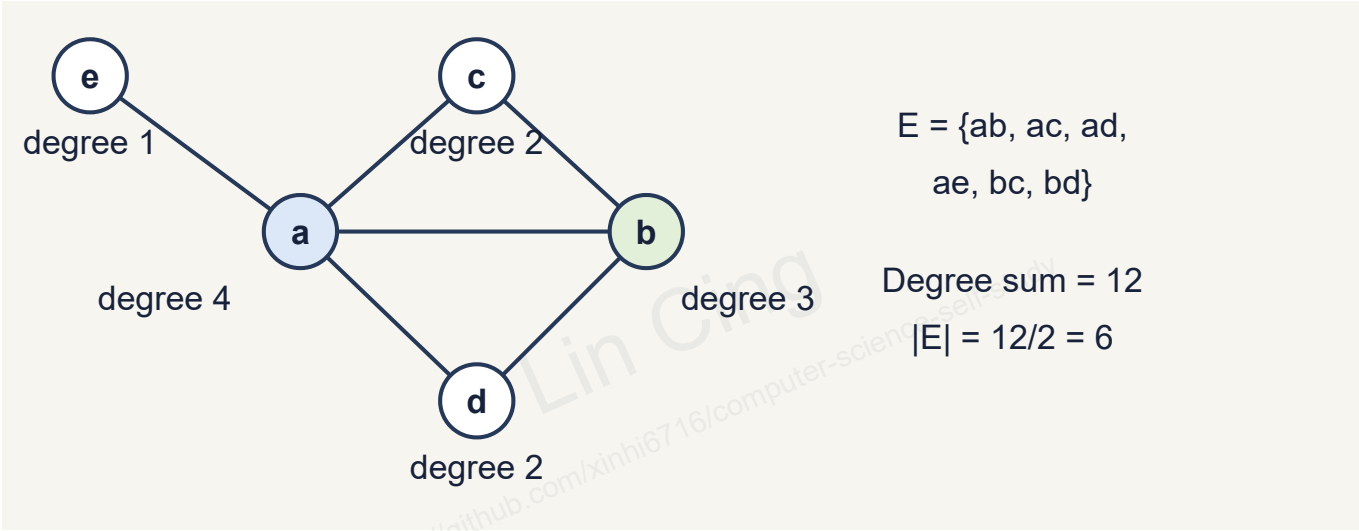
# Graph-theory solutions

## 11-E01 Odd-degree vertex count

The degree sum is twice the edge count and thus even. Even-degree vertices contribute an even sum, so odd-degree vertices also have an even sum. A sum of s odd integers has parity s; therefore s is even.

## 11-E02 Realize a degree sequence

The sum is twelve, so six edges are required. Havel–Hakimi first connects degree-four vertex to all four others, leaving sorted residual degrees (2,1,1,0). Connect degree two to the two positive degrees, leaving zeros. Explicit vertices a, b, c, d, e and edges ab, ac, ad, ae, bc, bd give degrees 4,3,2,2,1.




## 11-E03 Degrees and Euler traversal

The sum is 2・3+6・4=30=2e, giving e=15. Connectivity and exactly two odd vertices guarantee an Euler trail with those degree-three endpoints. There is no Euler circuit.

## 11-E04 Prove nonplanarity

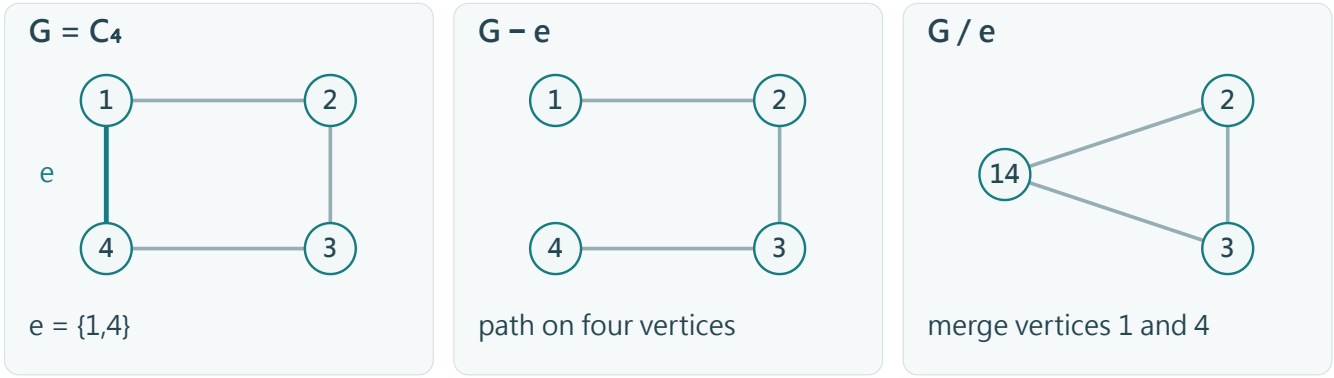
$K_{3,3}$ is simple bipartite and has no cycle shorter than four. A planar embedding would therefore satisfy $4f \le 2e$, counting each edge side once. It is connected, so Euler's formula with v=6,e=9 gives f=5. But $4f=20>18=2e$, contradiction. This excludes every embedding.

## 11-E05 Colorings of C4

Name cyclic vertices $v_1,v_2,v_3,v_4$ and split by opposite $v_1,v_3$ colors. Equal colors give k choices for that color and k−1 choices independently for each of $v_2,v_4$: $k(k-1)^2$. Distinct colors give $k(k-1)$ ordered choices and $(k-2)^2$ for the others. Hence

$$P_{C_4}(k) = k(k-1)^2 + k(k-1)(k-2)^2 = k(k-1)(k^2-3k+3).$$

At k=2 it gives two alternating colorings, as expected for this connected bipartite graph.




## 11-E06 Bridges and Hamilton conditions

Deleting a bridge separates two sides. A cycle using it would need another edge crossing back between the sides, contradicting that it is the unique cut edge. Thus no Hamilton cycle uses a bridge.

After connecting two $K_5$ graphs, n=10. Bridge endpoints have degree five, but the other eight vertices have degree four<5 and fail Dirac's universal threshold. In each clique choose a Hamilton path ending at its bridge endpoint; concatenate across the bridge to visit all ten vertices once. A Hamilton cycle would need to cross between cliques twice, impossible with the only crossing edge being that bridge. This is not a counterexample to Dirac; its degree premise fails.

## 11-E07 Embeddings and Euler formula

Both have V={A, B, C, D}, E={AB, AC, AD, BC, BD, CD}. The crossing of AC, BD is not in V and does not split either edge, so there remain four vertices and six edges. The crossed drawing alone proves nothing about nonplanarity, while the right crossing-free drawing proves $K_4$ planar.

Initially (v, e, f)=(4,6,4). AD is not a bridge because A–B–D remains; deleting it merges $F_1$,$F_3$, giving (4,5,3). BD is then not a bridge because B–C–D remains; deletion merges inner faces, giving (4,4,2). AC still has alternate A–B–C, so deletion gives (4,3,1) by merging inner and outer faces. Each step preserves v−e+f=2. Remaining AB, BC, CD form a tree. CD is now a bridge; deleting it would isolate D and cannot use the two-face merge argument.

For $K_3 \sqcup K_2$, v=5,e=4,c=2. The triangle encloses one face and all components share the unbounded face, so f=2. Then 5−4+2=3=1+c. Separate outer-face counts would duplicate one geometric region.

## 11-E08 Directed Euler connectivity

In A, B, C, D order both outdegrees and indegrees are 2,1,2,1. The nonisolated underlying graph is connected and every vertex balanced. A circuit is A→B→C→A→C→D→A, using AB, BC, CA, AC, CD, DA once each. There is no open Euler trail with different endpoints, since no vertices have imbalances +1,−1.

Adding X→Y→Z→X leaves old degrees unchanged and gives each new vertex indegree/outdegree one. Balance remains but nonisolated underlying connectivity fails: two components cannot be covered by one trail. All nine arcs cannot share a circuit. Both degree and connectivity conditions are essential.

## 11-E09 Nonplanarity of a periodic grid

There are 6 · 7=42 vertices. Each has four distinct neighbors (i±1,j),(i, j±1), giving degree four and e=42 · 4/2=84.

Every step changes exactly one coordinate. In a three-step closed walk changing both coordinates, one coordinate changes exactly once; ±1 cannot vanish modulo six or seven. If all three steps change one coordinate, the total change belongs to {−3,−1,1,3}, again nonzero modulo either. Thus no triangle exists. The graph is simple and connected, since coordinates can be adjusted consecutively to reach any vertex.

A triangle-free planar graph would have e≤2v−4=80, contradicting 84. The graph actually has a seven-cycle in the second-coordinate direction and is not bipartite. The proof uses triangle-freeness, not a false claim that every cycle is even.

## 11-E10 Three counts on labeled vertices

Five vertices have ten possible edges, so exactly four gives $\binom{10}{4} = 210$. Inclusion–exclusion for no isolates gives

$$2^{10} - 5 \cdot 2^6 + 10 \cdot 2^3 - 10 \cdot 2 + 5 - 1 = 1024 - 320 + 80 - 20 + 5 - 1 = 768.$$

For connectivity use $c_1$=1,$c_2$=1,$c_3$=4,$c_4$=38 and classify the component of vertex one. Disconnected contributions at k=1,2,3,4 are

$$\binom{4}{0} \cdot 1 \cdot 2^6 = 64, \quad \binom{4}{1} \cdot 1 \cdot 2^3 = 32, \quad \binom{4}{2} \cdot 4 \cdot 2 = 48, \quad \binom{4}{3} \cdot 38 = 152.$$

Thus $c_5$=1024−(64+32+48+152)=728. The difference forty consists of no-isolate disconnected graphs with component sizes 2+3. Choose the unique two-vertex component in ten ways; it has one edge, and the remaining three vertices have four connected graphs, giving 10 · 4=40. No division by two is needed because the different component sizes distinguish them uniquely.

## 11-E11 Vertex deletion and Hamilton paths

Removing |S| vertices from a Hamilton path leaves at most |S|+1 path segments. Other edges in G−S can only merge them, proving c(G−S)≤|S|+1, including empty S, which requires original connectivity.

For $K_{2,4}$ delete the two-vertex part. Four isolates remain, violating 4≤3, so no Hamilton path exists. Equivalently an alternating path cannot use part sizes differing by more than one.

$K_{3,4}$ has path $b_1$–$a_1$–$b_2$–$a_2$–$b_3$–$a_3$–$b_4$. Every edge crosses parts and each vertex occurs once. A cycle would alternate equally many vertices from both parts, impossible for 3≠4. Alternatively deleting the three-vertex part leaves four components, violating the Hamilton-cycle bound.

## 11-E12 Colorings of C5

Sequential coloring gives P_$C_3$(k)=k(k−1)(k−2). Deleting an edge of $C_4$ gives a four-vertex tree; contracting it gives $C_3$:

$$P_{C_4} = k(k-1)^3 - k(k-1)(k-2) = k^4 - 4k^3 + 6k^2 - 3k.$$

For $C_5$ deletion gives a five-vertex tree and contraction $C_4$:

$$P_{C_5} = k(k-1)^4 - P_{C_4} = k^5 - 5k^4 + 10k^3 - 10k^2 + 4k.$$

At two, 32−80+80−40+8=0: an odd cycle is not two-colorable. At three, 243−405+270−90+12=30 proper colorings with three named colors exist. Every one actually uses all three, since using fewer would give a two-coloring. Thus χ($C_5$)=3; thirty counts colorings, not the minimum number of colors.

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## 11-E13 Trace stack-based Euler traversal

Each adjacency list stores IDs in order: A:[1,3], B:[1,2], C:[2,3,4,6], D:[4,5], E:[5,6]. Initially S=[A], answer=[]. The complete trace is:

| Action | S (top at right) | answer (append order) |
|---|---|---|
| e1 | `[A,B]` | `[]` |
| e2 | `[A,B,C]` | `[]` |
| e3 | `[A,B,C,A]` | `[]` |
| pop A | `[A,B,C]` | `[A]` |
| e4 | `[A,B,C,D]` | `[A]` |
| e5 | `[A,B,C,D,E]` | `[A]` |
| e6 | `[A,B,C,D,E,C]` | `[A]` |
| pop C | `[A,B,C,D,E]` | `[A,C]` |
| pop E | `[A,B,C,D]` | `[A,C,E]` |
| pop D | `[A,B,C]` | `[A,C,E,D]` |
| pop C | `[A,B]` | `[A,C,E,D,C]` |
| pop B | `[A]` | `[A,C,E,D,C,B]` |
| pop A | `[]` | `[A,C,E,D,C,B,A]` |

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The first pop removes only the final A frame. C remains on the stack and has unused edge e4, so the search continues. Reversing the final answer gives A–B–C–D–E–C–A and IDs 1,2,4,5,6,3. Each ID occurs once, including e3 at the end.

After replacing EC by EF, degrees are A:2, B:2, C:3, D:2, E:2, F:1. The nonisolated part is connected, so an open Euler trail exists, but its endpoints must be C and F. Starting at A violates the start condition. With that start, the algorithm produces A–B–C–D–E–F–A after reversal: the final pair F–A is not an edge. A count of m used IDs and m+1 output vertices alone does not guarantee a valid route when the preconditions are omitted. Starting at C instead gives C–B–A–C–D–E–F with IDs 2,1,3,4,5,6.

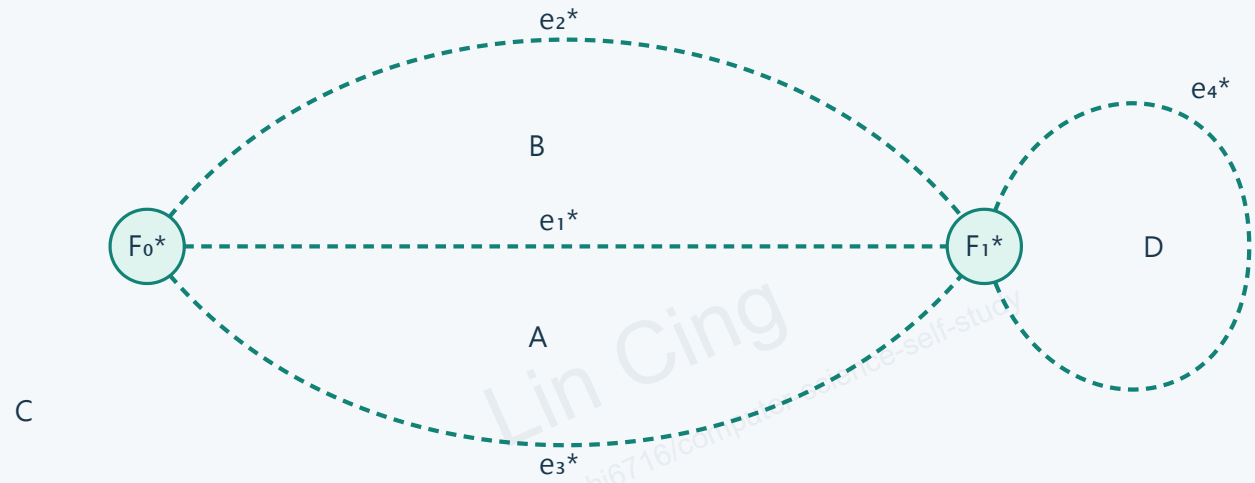
## 11-E14 Bridges and plane duals

### Use the same embedding for boundaries and dual incidences

The exterior walk A→C→B→A has length 3; the interior walk A→B→C→D→C→A has length 5. CD occurs twice in the interior boundary, so the total is 8, twice the edge count. These are not the lengths of two simple cycles.

Edges $e_1^*, e_2^*, e_3^*$ each join $F_0^*, F_1^*$, but have distinct IDs. Edge $e_4^*$ is a loop at $F_1^*$. Degrees are 3 and $3 + 2 = 5$. The connected dual has two vertices and four edges, hence $f^* = 2 - 2 + 4 = 4$. A, B, C, D in the drawing label faces; only the two circles are dual vertices.

#### The dual alone: three parallel edges and one loop




A, B, C, D label four dual faces. A loop contributes two to degree.

### Restart from the original four-edge graph for each deletion

Deleting AB leaves tree AC, BC, CD, so $(v, e, f, c) = (4, 3, 1, 1)$ and $4 - 3 + 1 = 2 = 1 + c$. Deleting CD leaves the triangle and isolated D, so $(v, e, f, c) = (4, 3, 2, 2)$ and $4 - 3 + 2 = 3 = 1 + c$. D still counts toward v and c but encloses no new face.

If D is moved outside, both sides of CD belong to the exterior and the loop attaches to $F_0^*$. Primal counts remain four vertices, four edges, two faces; dual counts remain two vertices, four edges, four faces. The named-face degrees become $5, 3$. The two abstract duals need not be nonisomorphic: swapping the face vertices gives an isomorphism here. What changes is which named face carries the loop in the embedding.

## 11-E15 Greedy order and the true chromatic number

| Order | Vertex | Prior-neighbor colors | Chosen color |
|---|---|---|---|
| $1, 4, 2, 3$ | 1 | $\varnothing$ | 1 |
| | 4 | $\varnothing$ | 1 |
| | 2 | $\{1\}$ | 2 |
| | 3 | $\{1, 2\}$ | 3 |
| $1, 2, 3, 4$ | 1 | $\varnothing$ | 1 |
| | 2 | $\{1\}$ | 2 |
| | 3 | $\{2\}$ | 1 |
| | 4 | $\{1\}$ | 2 |

In the first run vertex 3 sees differently colored neighbors 2 and 4. It must choose color 3 under the earlier decisions. Yet vertices 2 and 4 are nonadjacent and could share a color; greedy never revisits their assignments.

An edge gives $\chi \geq 2$; the second run supplies a two-coloring, giving $\chi \leq 2$. Thus $\chi = 2$ and the graph is bipartite. A three-color greedy run supplies only an upper bound, not a lower bound.

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## 11-E16 Matrix powers and isomorphism invariants

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

$(A^2)_{22} = A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} = 1 + 0 + 1 = 2$. Its walks are $2 - 1 - 2$ and $2 - 3 - 2$, neither a simple path.

| Invariant | $C_6$ | $C_3 \sqcup C_3$ |
|---|---|---|
| Vertices | 6 | 6 |
| Edges | 6 | 6 |
| Degree sequence | $(2, 2, 2, 2, 2, 2)$ | $(2, 2, 2, 2, 2, 2)$ |
| Triangles | 0 | 2 |
| Components | 1 | 2 |

Either of the last two differences disproves isomorphism. Agreement on the first three does not prove it.

## References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study